Tanmay Bhuskute

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SUMMARY

Backend Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience building Python backend services, REST APIs, and PostgreSQL-backed data layers on AWS and Kubernetes. Current research integrates LLM and computer vision workflows into automated pipelines.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, C/C++, SQL, Go.

Backend & Web: REST APIs, GraphQL, Django, Flask, FastAPI, Node.js, React, HTML/CSS.

Databases: PostgreSQL, MySQL, RDBMS, Redis.

DevOps & Cloud: Git, Linux/Unix, Docker, Kubernetes, Infrastructure as Code (Terraform, Ansible), AWS, Azure, Google Cloud Platform, Incident Response, Computer Networking.

ML & Computer Vision: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, OpenCV, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Built two-stage memory-augmented LLM pipeline in Python integrating computer vision grounding for Android GUI bug reproduction, benchmarked against a GPT-4o + CLIP baseline across 53 runs on 20+ apps.
- Achieved **25.7x latency reduction** (39.4s vs 1012.1s avg per run) in production-style inference pipeline while preserving task-completion accuracy.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Built Python (Boto3) backend automation for Teamcenter X infrastructure services, replacing legacy Ruby scripts and cutting deployment time **40%** across AWS environments.
- Diagnosed and resolved production incidents across **44 environment upgrades** in 4 months, debugging live services on Linux, EC2, databases, and internal networking.
- Designed Ansible-based Blue-Green deployment framework, reducing production incident downtime by **50%** through faster rollback.
- Migrated **40+ production services** to Kubernetes with ArgoCD, Helm, and GitLab CI/CD, building infrastructure-as-code workflows for repeatable deployment.
- Served as Quality Coach and SonarQube Specialist, raising automated test coverage **35% to 78%**; mentored **5 engineers** on code review and CI/CD practices.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Built **Go + Terratest** testing framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, FastAPI, PostgreSQL, Redis, Docker] Jul 2026

- Built ML backend forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** dataset); served real-time predictions via REST API backed by PostgreSQL and Redis caching, containerized with Docker.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).